125101 - Analytical chemistry 1 for engineers

Tutorial 12 – Gravimetric Analysis

T12-1. A 0.649 g sample containing K_2SO_4 ($M.W$ 174.27 g/mol) and $(NH_4)_2SO_4$ ($M.W$ 132.14 g/mol) was dissolved in water and treated with $Ba(NO_3)_2$ ($M.W$ 233.39 g/mol) to precipitate all SO_4^{2-} as $BaSO_4$. Find the weight percent of K_2SO_4 in the sample if 0.977g of precipitate was obtained.

T12-2. In gravimetric analysis, an organic compound with a formula mass of 417 g/mol was analyzed for ethoxyl (CH_3CH_2O-) group by the reactions:

$$CH_3CH_2OR + HI \longrightarrow ROH + CH_3CH_2I$$

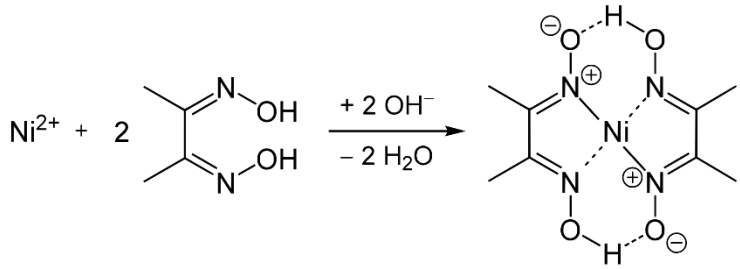
$$CH_3CH_2I + Ag^+ + OH^- \longrightarrow AgI(\downarrow) + CH_3CH_2OH$$

A 25.42 mg sample of the compound produced 29.03 mg of AgI. How many ethoxyl groups are there in each molecule?

T12-3. To measure the Nickel (Ni) content in steel, the alloy is dissolved in 12 M HCl and neutralized in the presence of citrate ion. The slightly basic solution is warmed, and alcoholic dimethylglyoxime (DMG) is added to precipitate the red $DMG{-}Ni$ complex quantitatively. The product is filtered, washed with cold water, and dried at 110°C.

a. If the Ni content is known to be near 3 wt% and you wish to analyze 1.0 g of steel. What volume of 1.0 wt% alcohol DMG solution should you use to give a 50% excess of DMG for the analysis? The density of DMG is 0.79 g/mL.

b. If 1.1634 g of steel gives 0.1795 g of precipitate, what percentage of Ni is in the steel?



M.W (Ni) = 58.69 g/mol

M.W (DMG) = 116.12 g/mol

M.W (Complex) = 288.91 g/mol

T12-4. Precipitate's liquids are liquid that stay with the precipitate and contain contaminants. If the volume of the precipitate liquid is 1mL, and the washing liquid is 20 mL, please try to calculate the contaminants in the precipitate's liquid after n washing cycles, n = 1, 2, 3, respectively. How will your answer change if the washing liquid would increase to 60 mL? What are the conclusions from your results?